Circle patterns and primitive fillings of augmented links

Abstract

A simple triangulated sphere with a matching of its dual determines a link diagram with specified crossing circles. We construct the complete hyperbolic exterior, deform one common marked circle pattern, and calculate the original peripheral words. An explicitly normalized classical action then has a critical point for all independently sufficiently long signed fillings. Its real critical value, multiplied by 2π, is the volume of the smoothly filled original exterior. The proof includes the logarithmic torus periods and the absolute fundamental-chain comparison; an exponentiated filling equation alone would not imply either conclusion. No quantum asymptotic is used. The final section states the precise interface with the separate oscillatory filling theorem.

1 The diagram, the action, and the theorem

An *augmented diagram* consists of strands in a projection sphere, with disjoint two-strand crossing boxes. Around each box is an auxiliary unknotted component, its *crossing circle*, bounding a disk pierced twice by the strands. Removing full twists leaves either no crossing or one signed half twist in each box. The disk and its collar are retained: they specify the peripheral curves used to put the full twists back.

Here is the combinatorial input throughout this paper. Let $\mathcal{T}$ be a finite oriented simple triangulated sphere with at least four vertices. Choose a perfect matching of its cubic dual. Two matched triangles share one edge and have distinct third vertices. Write their cyclic quadrilateral as (x_e, b_e, y_e, c_e); its diagonal is $x_e y_e$. The triangles are the input (x_e, b_e, y_e) and output (x_e, c_e, y_e), with their sphere orientations understood. There are R matched pairs and $R+2$ white vertices. An edge of $\mathcal{T}$ whose dual is not in the matching will be called a *strand edge*. Choose such an edge uv as a mark.

The inverse cut in Section 2 constructs from these data an augmented link in S^3. Equivalently, the input may be an actual augmented diagram whose projection and disk cut has exactly these labels and orders. At each disk retain $\epsilon_e \in \{-1, 0, 1\}$ half twists, with the positive convention specified by the Hurwitz move in (4.7). Denote the resulting augmented exterior by Y_D. Its main components are the original strands; its auxiliary components are the crossing circles.

On each auxiliary cusp choose the original disk-framed primitive meridian and longitude (μ_e, λ_e). The meridian goes once around the crossing circle. The longitude follows its twice-punctured disk and is represented by the ordered product $P_e = B_e A_e$ of the two based main meridians. The orientation and basepaths will be checked in Section 4. Filling the slope $\mu_e + p_e\lambda_e$, $p_e \in \mathbb{Z}$, inserts $-2p_e$ crossings, so the restored power is

$$n_e = \epsilon_e - 2p_e. \tag{1.1}$$

The original outside pairing and these box parities determine the main components. We call the filled link a knot only when that terminal graph is connected.

We use curvatures, not radii, as the variables of the action. A finite white circle with curvature $w > 0$ has radius $1/w$. The two marked white circles will be parallel lines and have

curvature zero. For three nonnegative curvatures, not all zero, define

$$\begin{aligned} F(w_1,w_2,w_3) &= \frac{1}{2\pi} \max_{\substack{\theta_j\ge 0\\ \sum_j \theta_j=\pi}} \left\{ \sum_j \theta_j \log w_j + \mathcal{V}(\theta) \right\}, \\ \mathcal{V}(\theta) &= 2\sum_{j=1}^{3} \left[\Lambda(\theta_j/2) + \Lambda((\pi-\theta_j)/2) \right], \\ \Lambda(t) &= -\int_0^t \log|2\sin s|\, ds. \end{aligned} \tag{1.2}$$

A zero curvature forces its angle to zero; its logarithmic term is omitted. Thus $F(w,0,0) = \frac{1}{2}\log w$. This definition fixes the absolute constant geometrically.

For one triangle (x,b,y), introduce a small complex parameter δ, and put

$$\begin{aligned} &s = \sin(\pi\delta), && c_0 = \cos(\pi\delta), && A = x+y, \\ &D = A^2 - 4xys^2, && T_b = \sqrt{bA + xy + b^2 s^2}, && E_b = A + 2bs^2 + 2sT_b. \end{aligned} \tag{1.3}$$

The square roots are continued from positive complete values. In a singleton triangle, where a middle and an outer curvature are zero, set $T_b = 0$ and $E_b = A$ exactly. At every matched pair $A > 0$. Define the holomorphic triangle germ Q_b by

$$Q_b(0;x,b,y) = F(x,b,y), \qquad \partial_\delta Q_b = -\operatorname{Log}(E_b/\sqrt{D}), \tag{1.4}$$

with logarithm zero at $\delta = 0$. Section 3 proves that it is an angle potential for the deformed common pattern. It has logarithmic homogeneity $1/2$, as does F.

Set $w_u = w_v = 0$, fix one common scale of the ordinary curvatures, and define the selected face-critical value

$$A_0(\boldsymbol{\delta}) = \operatorname{crit}_w \left\{ \sum_{e=1}^{R} (Q_{b_e} + Q_{c_e}) - \sum_{f\ne u,v} \operatorname{Log} w_f \right\}. \tag{1.5}$$

The critical point in this definition is the unique germ at the complete positive packing. Its existence is proved below.

Theorem 1.1 (Primitive fillings and their classical action)**.** *For every fixed marked matched sphere and its retained signed collars, the action (1.5) is holomorphic near zero, separately even, and real on real parameters. Its complete value is*

$$A_0(0) = \Phi_D = \frac{\operatorname{Vol}(Y_D)}{2\pi}.$$

There is a diagram-dependent threshold T_D with the following property. For every fixed integer vector $\mathbf{p}$ with $\min_e |n_e| > T_D$, the function

$$S_{\mathbf{n}}(\delta) = A_0(\delta) + \pi\mathrm{i}\sum_e n_e\delta_e^2 + 2\pi\mathrm{i}\sum_e \delta_e \tag{1.6}$$

has one critical point in the selected small germ. At that point:

(i) The explicitly constructed representation of $\pi_1(Y_D)$ has original primitive similarity logarithms

$$U_e = -\operatorname{Log}(E_{b_e}E_{c_e}/D_e) + 2\pi\mathrm{i}\epsilon_e\delta_e, \qquad V_e = -4\pi\mathrm{i}\delta_e, \qquad U_e + p_eV_e = -2\pi\mathrm{i}. \tag{1.7}$$

Moreover $\operatorname{Im}\delta_e > 0$, for either sign of n_e.

(ii) It is the holonomy of a smooth complete finite-volume hyperbolic metric on the original filled exterior $Y_D(\mu_e + p_e\lambda_e)_e$. Every main cusp remains complete.

(iii) Its critical value satisfies the absolute equality

$$2\pi \operatorname{Re} S_{\mathbf{n}} = \operatorname{Vol}(Y_D(\mu_e + p_e\lambda_e)_e). \tag{1.8}$$

If the original component test is a knot, the right side is $\operatorname{Vol}(S^3 \setminus K_{\mathbf{p}})$. *As all* $|n_e|$ *tend to infinity, the critical point tends to zero and* $\operatorname{Re} S_{\mathbf{n}} \to \Phi_D$.

The sphere, its matching, collars and filling integers are fixed before any asymptotic application. The threshold places no restriction on the ratios or signs of the independently long powers. The uniqueness claim concerns the indicated critical germ; it is not uniqueness of the whole character variety. No quantum limit occurs in the theorem.

Variational existence and uniqueness for circle patterns, and their relation to hyperbolic volume, are classical; see Rivin [5] and Bobenko–Springborn [1]. We give the particular packing and deformation arguments needed here, including the marked lines and the original peripheral paths. We use the elementary holomorphic implicit-function theorem, differential forms and Stokes; for the smooth background see [2, Chapters 4 and 16]. The implicit-function arguments may equivalently be obtained by preconditioning the equation with its invertible linear part and applying uniform contraction on a small complex ball. The resulting fixed point is holomorphic as the locally uniform limit of its holomorphic iterates. The ideal-tetrahedron volume formula used below is the classical Lobachevsky formula; see [3, Appendix]. We prove the global normalizations and filling comparison rather than importing a general Dehn filling or character-rigidity theorem.

2 A packing and the original complete exterior

2.1 Strict feasibility and the circle packing

Proposition 2.1 (Packing with its exact labels)**.** *Every simple triangulated sphere in the hypotheses has a packing of white disks, with contacts exactly its edges. The circles through the three contacts of each triangular interstice bound orthogonal shaded planes. These planes and the white planes bound a convex ideal right-angled polyhedron with precisely the prescribed medial sphere.*

Proof. Remove a triangular face abc, and prescribe positive boundary radii r_a, r_b, r_c. The remaining disk, with v interior vertices, has $2v + 1$ triangles by Euler's formula. Give each triangle angle supply π, each interior vertex demand 2π, and the three boundary vertices pooled demand π; their separate corners are retained.

We first find a strictly positive allocation to all corners. A nonempty set S of interior vertices touches at least $2|S| + 1$ disk triangles. Indeed, puncture the sphere at a vertex of S. On the complementary interior vertices T and a, b, c, a simple planar graph has at most $2|T| + 1$ bounded triangular faces. One is the omitted face abc, so at most $2|T|$ disk triangles avoid S. A proper destination set containing the pooled boundary destination also has strict slack: its complement T consists only of interior vertices, and at most $2|T| - 5$ triangles are wholly supported there for $|T| \geq 3$, with none for $|T| \leq 2$. The full destination set has equal demand and supply.

These are the finite transport Hall inequalities. To recall the short argument, join a source to the triangle nodes with their supply capacities, join each triangle to its incident destinations, and join destinations to a sink with their demands. A finite cut minimizes to one of these inequalities. Augmenting along residual paths proves feasibility for rational

capacities; compactness gives the real case. Strict slack at every proper destination subset permits subtraction of one small positive amount at every corner before this argument. Thus an allocation positive at every corner exists.

On the compact angle polytope maximize

$$\sum_t \mathcal{V}(\theta_t) + \sum_{\text{boundary corners } tf} \theta_{tf} \log(1/r_f).$$

For $0 < \theta < \pi$, the corner derivative of $\mathcal{V}$ is $-\log\tan(\theta/2)$, and its second derivative is $-1/\sin\theta < 0$. A maximum is interior: moving toward the strictly positive allocation gives a positive $\varepsilon\log(1/\varepsilon)$ contribution at each new zero corner, a nonnegative one at an old π corner, and only bounded linear costs elsewhere. Strict feasibility alone would not establish this boundary exclusion. Strict concavity now makes the maximum unique. Its Lagrange multipliers give positive curvatures $w_f = 1/r_f$, including the prescribed boundary ones, and

$$\tan(\theta_{tf}/2) = \frac{w_f}{D_t}, \qquad D_t = \sqrt{\sum_{\{f,g\}\subset t} w_f w_g}. \tag{2.1}$$

The row sum π determines D_t by the tangent addition identity. These are the angles of the Euclidean center triangle with sides $r_f + r_g$.

Glue those center triangles. Interior angle sums are 2π. Continuation around an interior vertex fixes that vertex and rotates by 2π, hence has zero translation as well as rotation. Such loops generate the punctured disk's holonomy, so the development is globally defined. Its boundary consists of the three prescribed straight sides; each boundary angle sum is positive and less than π, and their total is π. It therefore traverses a convex triangle once. The development is locally orientation-preserving at every interior point. For a point off the developed edges the number of preimages is its boundary winding number, one inside and zero outside. Local injectivity extends the conclusion across the edges and vertices. Thus the development is an embedding.

Place each radius-r_f disk at its developed center. In a center triangle the opposite edge is covered by the other two tangent disks, both disjoint from the third. Their mutual contact is strictly outside the third disk, whose contacts with them lie on the two incident edges. Consequently the third disk's sector stops strictly before the opposite edge. Sectors assemble to full disks at interior vertices and to the prescribed boundary sectors. This proves disjoint interiors and excludes unintended contacts, even across separating triangles.

The circle through a triangle's three contacts is its incircle: the contact distance along a center edge is the corresponding radius. It is orthogonal to each vertex circle. Its disk lies in its center triangle and contains no other packing contact. For the omitted outer triangle choose the opposite side of its contact circle; all interior contacts lie in its bounded curvilinear gap and therefore inside that circle. Each contact consequently lies in all prescribed inward white and shaded halfspaces, strictly unless incident. The exposed faces form the closed labeled medial sphere, so the convex hull has exactly those facets. Four faces alternate at every contact, with successive right angles. This constructs the stated polyhedron Q. □

Sending the contact of the marked edge uv to infinity by one orientation-preserving Möbius map makes its white disks the outward halfplanes $Y \le 0$ and $Y \ge 1$. All other disks are bounded in the strip. The curvature scale is immaterial. In particular exactly the two marked curvatures vanish.

2.2 The reversible cut and primitive cusps

Take disjoint small disk neighborhoods of the matched dual edges. They are four-terminal disks. At disk e, write the cyclic terminals C_0, C_1, C_2, C_3. The two shaded halves group

(C_1, C_2) and (C_3, C_0), while the untwisted physical strands join (C_0, C_1) and (C_2, C_3). Join the outside terminals along the nonmatching dual edges, and place a perpendicular crossing circle and its disk in each disjoint three-ball collar. This is an embedded link in S^3. For odd parity the physical pairing is $(C_0, C_2), (C_1, C_3)$. The outside and inside terminal matchings form a degree-two graph; its connectedness is exactly the knot test. Full twists do not change it.

Cut the augmented exterior along the projection sphere. In each projection ball every crossing-circle semicircle is boundary parallel by its original half disk. Drilling that semicircle and opening that same half disk cancels the introduced handle. The half disks are disjoint, so all these operations leave a ball in each copy. Its boundary is the labeled medial sphere of Proposition 2.1. The boundary cell identification with Q extends radially over the ball. Cycles in the matching graph introduce no extra handle.

Use two copies Q^+, Q^-. White faces glue across copies by their projection labels. At an even disk, in each copy, the shaded pairing fixes the crossing-circle vertex O_e and sends

$$C_1 \longmapsto C_0, \qquad C_2 \longmapsto C_3. \tag{2.2}$$

At an odd disk it interchanges copies and punctures:

$$\begin{aligned} (O, C_1, C_2)^+ &\longmapsto (O, C_3, C_0)^-, \\ (O, C_3, C_0)^+ &\longmapsto (O, C_1, C_2)^-. \end{aligned} \tag{2.3}$$

These maps follow by rotating one side of the twice-punctured disk by $\pm\pi$, which sends its normal coordinates (y, z) to $(-y, -z)$. The two signs have the same corner permutation but different collar paths. Those paths will remain in the peripheral calculation.

Every ideal vertex link of Q is a rectangle. Choose its horosphere so its two opposite shaded sides have length one. Thus every shaded ideal triangle has its canonical length-one horocycles at its three corners. The prescribed triangle maps, including odd permutations, preserve them, as do the white identifications. Each edge cycle has four pieces and angle 2π. It fixes the two ideal endpoints, preserves both horospheres, and therefore has zero translation along the edge as well as rotation 2π; its full holonomy is the identity. The links glue to the original torus links without cone or dilation defects. Their horoball ends are complete, and the remaining part is compact. We have proved

$$\operatorname{Vol}(Y_D) = 2 \operatorname{Vol}(Q). \tag{2.4}$$

This quotient also has a positive ideal triangulation. Choose a vertex of Q, triangulate its nonincident faces, and cone their triangles to that vertex; fan the incident faces. Strict support makes every tetrahedron nondegenerate and positive. Use identical white subdivisions in the two copies; shaded faces were already triangles. These are cells of this exterior, not an assumed triangulation of a later filled manifold.

2.3 The absolute complete action

At the complete marked packing, let θ_{tf} be the center angles, with the usual line limits. A singleton triangle has angles $(\pi, 0, 0)$. Formula (2.1) and strict concavity prove

$$2\pi F(w_t) = \mathcal{V}(\theta_t) + \sum_{f \in t} \theta_{tf} \log w_f, \qquad \partial_{\log w_f} F = \theta_{tf}/(2\pi).$$

The finite-circle angle sums are 2π, so the complete packing is stationary for $\sum_t F(w_t) - \sum_{f \neq u,v} \log w_f$.

To fix its absolute value, cone the marked ideal vertex, now infinity, to every nonincident facet of Q. These positive pyramids partition Q. For a cyclic white ideal polygon with consecutive central arcs α_j, its cone volume is $\sum_j \Lambda(\alpha_j/2)$. Triangulate the polygon: the ideal tetrahedron formula gives three Lobachevsky values, and internal diagonal half-arcs cancel by $\Lambda(\pi - t) = -\Lambda(t)$. At a shaded triangle the white contact arcs are θ_{tf}. Orthogonality gives complementary shaded arcs $\pi - \theta_{tf}$. Consequently

$$\mathrm{Vol}(Q) = \sum_{t, f\in t} [\Lambda(\theta_{tf}/2) + \Lambda((\pi - \theta_{tf})/2)] = \frac{1}{2}\sum_t \mathcal{V}(\theta_t).$$

Incident line facets and singleton triangles contribute zero by continuity. Canceling the angle-weight terms against the original face targets proves

$$\Phi_D = \sum_t F(w_t) - \sum_{f\neq u,v} \log w_f = \mathrm{Vol}(Q)/\pi = \mathrm{Vol}(Y_D)/(2\pi). \tag{2.5}$$

The constant $F(w, 0, 0) = \frac{1}{2}\log w$ is essential here.

3 One holomorphic marked pattern

Use the strip $0 < Y < 1$. For each ordinary circle retain its center (X_f, Y_f) and radius r_f. A strand edge has inversive cosine $I_{fg} = 1$, and a matching edge e has $I_{fg} = \cos(2\pi\delta_e)$. The actual signed equations are

$$\begin{aligned} (X_f - X_g)^2 + (Y_f - Y_g)^2 &= r_f^2 + r_g^2 + 2I_{fg} r_f r_g, \\ Y_f &= r_f I_{uf} && \text{on the lower line,} \\ 1 - Y_f &= r_f I_{vf} && \text{on the upper line.} \end{aligned} \tag{3.1}$$

The line-line edge is automatic. Fix one finite center's X coordinate. There are $3R - 1$ equations and $3R - 1$ variables. Complexification treats X, Y, r as independent holomorphic coordinates; it does not conjugate the deformation parameters.

Lemma 3.1 (The complete pattern Jacobian)**.** *The Jacobian of* (3.1) *in these normalized variables is invertible at completeness.*

Proof. For the ordinary corners I_t of one triangle put $q_i = \log w_i$, $D_t^2 = \sum_{i<j} w_i w_j$. Direct differentiation of (2.1) gives

$$\begin{aligned} \partial_{q_i} F_t &= \pi^{-1} \arctan(w_i/D_t), \\ \partial_{q_i q_j} F_t &= -\kappa_{ij}^t, \qquad \kappa_{ij}^t = \frac{w_i w_j}{2\pi D_t (w_i + w_j)} > 0 \quad (i \neq j), \\ h^{\mathsf{T}} \nabla^2 F_t h &= \sum_{i<j} \kappa_{ij}^t (h_i - h_j)^2. \end{aligned} \tag{3.2}$$

For three corners use $D_t^2 + w_i^2 = (w_i + w_j)(w_i + w_k)$; two corners are its zero-curvature limit. Homogeneity gives zero row sums. A singleton has potential $q_i/2$ and zero Hessian.

Deleting u, v leaves a connected ordinary graph. Reroute any path through u along its cyclic link; then reroute through v along its link with u removed, which is a connected path. Every surviving edge contributes positive conductance in (3.2). Hence the full Hessian has precisely the constant kernel.

An infinitesimal solution of the circle equations preserves the cyclic order of the center directions. Their angle sums at finite centers remain 2π, so the Hessian calculation makes

every logarithmic radius variation the same scale. The two triangles containing uv have a circle tangent to both lines. Its complete radius is $1/2$; linearizing its two line equations gives $dr = 0$. Thus every radius variation is zero.

With radii fixed, each finite center triangle has one infinitesimal Euclidean motion. A one-line triangle has $dY_f = dY_g = 0$; its horizontal separation is $2\sqrt{r_f r_g} > 0$, and its fixed edge gives $dX_f = dX_g$. Its motion is horizontal translation. Delete the two triangles incident to uv. The remaining triangulated disk has connected triangle adjacency, contains every finite vertex, and contains a one-line triangle. Across a finite edge, agreement at two distinct centers identifies the motions. Across a line edge, agreement at the common finite center identifies the two horizontal translations. All centers therefore have the same horizontal translation. The fixed X coordinate removes it. The square Jacobian has zero kernel. □

The holomorphic implicit-function theorem now gives a unique nearby pattern on one fixed polydisk. It is real for real parameters and even separately in each δ_e. Radii stay nonzero. Real cyclic orders and main tangencies persist.

We identify its angle potential explicitly. For one deformed triangle, the middle and outer angles satisfy

$$\begin{aligned} \theta_b &= 2\arctan(bc_0/T_b), \\ \cos\theta_x &= \frac{x(y-x)+b(y+x\cos(2\pi\delta))}{(x+b)\sqrt{D}}, \\ \sin\theta_x &= \frac{2xc_0T_b}{(x+b)\sqrt{D}}. \end{aligned} \tag{3.3}$$

Exchange x, y for the third angle. Differentiation gives

$$\partial_\delta(\theta_f/(2\pi)) = -\partial_{\log w_f}\operatorname{Log}(E_b/\sqrt{D}).$$

For the middle angle both sides are $-bs/T_b$. For an outer angle, differentiate its sine and cosine and use their squared sum one; substitution gives the same identity. At zero the derivatives are those of F, so integration with (1.4) proves

$$\partial_{\log w_f} Q_b = \theta_f/(2\pi). \tag{3.4}$$

One zero curvature is the line limit. With two zero curvatures the only finite angle is π, and the potential is $\frac{1}{2}\log w$.

The common pattern's angle sums prove the face-critical equations. The grounded Hessian (3.2) gives their unique normalized holomorphic solution, so it is exactly the germ used in (1.5). In particular, the two degenerate triangles at the mark do not choose their strip widths independently: a finite circle tangent to one line and with inversive angle $2\pi\delta_e$ at the other obeys

$$r_f(1+\cos(2\pi\delta_e)) = 1. \tag{3.5}$$

The circle equations, rather than a repeated scalar root, enforce this common strip.

The ordered product of outer reflections has trace $-2\cos(2\pi\delta_e)$. Its noncentral complete parabolic limit has a unit off-diagonal entry in some chart. The eigenvalues $J_e = -e^{-2\pi i\delta_e}$ and J_e^{-1} therefore specify holomorphic fixed-line germs: solve the linear eigenvector equation using that unit entry. The two flags may coincide at completeness; no division by their difference is needed.

4 The crossing disk and the original peripheral words

4.1 One primitive sidepairing

We calculate a map, not just its multiplier. Products act from right to left. Let $\mathcal{R}_f$ denote reflection in white circle f. The ordered input and output main meridians are

$$\begin{aligned} A_{\text{in}} &= \mathcal{R}_b\mathcal{R}_x, \quad B_{\text{in}} = \mathcal{R}_y\mathcal{R}_b, \quad P = B_{\text{in}}A_{\text{in}} = \mathcal{R}_y\mathcal{R}_x, \\ A_{\text{out}} &= \mathcal{R}_c\mathcal{R}_x, \quad B_{\text{out}} = \mathcal{R}_y\mathcal{R}_c. \end{aligned} \tag{4.1}$$

The actual contacts are $C_1 = x \cap b,\ C_2 = b \cap y$ and $C_0 = x \cap c,\ C_3 = c \cap y$. The pure disk pairs $C_1 \mapsto C_0,\ C_2 \mapsto C_3$.

First take positive finite curvatures and small real $\delta > 0$. Place the outer centers at $0, L$, where

$$L^2 = (r_x + r_y)^2 - 4r_xr_ys^2 = \frac{D}{x^2y^2}.$$

Their intersections are $P_\pm = u \pm \mathrm{i}v$, with

$$u = \frac{L^2 + r_x^2 - r_y^2}{2L}, \qquad v = \frac{2sc_0}{\sqrt{D}}.$$

The input middle center lies above the outer center line, the output below it. Their coordinates are

$$p_f = u + \frac{r_f(r_x - r_y)}{L}, \qquad q_b = \frac{2c_0T_b}{b\sqrt{D}}, \qquad q_c = -\frac{2c_0T_c}{c\sqrt{D}}.$$

Subtracting the three orthogonality equations shows that the shaded circle on side b has center $u + \mathrm{i}t_b$, where

$$t_b = \frac{c_0(A + 2bs^2)}{\sqrt{D}\,T_b}, \qquad \text{squared radius } t_b^2 - v^2.$$

On side c the analogous center has negative imaginary part. In the coordinate $w = (z - P_+)/(z - P_-)$, the outer circles become lines through zero and the shaded circles are concentric:

$$\rho_b^2 = \frac{D}{E_b^2}, \qquad \rho_c^2 = \frac{E_c^2}{D}, \qquad R_0 = \frac{\rho_c}{\rho_b} = \frac{E_bE_c}{D}. \tag{4.2}$$

Indeed a circle of center $u + \mathrm{i}t$ and squared radius $t^2 - v^2$ has constant $|w|^2 = (t - v)/(t + v)$; use $(A + 2bs^2)^2 - 4s^2T_b^2 = D$.

Both middle circles occupy the same wedge. Away from the outer tangency $w \to 1$ as $\delta \to 0$; both finite middle circles are separated from that tangency. A circle tangent to the two sides of a fixed wedge is determined by its positive size. Consequently $C_w(w) = R_0w$ is exactly the prescribed disk pairing. The product of the two shaded reflections instead has ratio R_0^2; it is not the primitive path.

A determinant-one lift of the outer product is

$$P = \frac{1}{r_xr_y}\begin{pmatrix} r_y^2 - L^2 & Lr_x^2 \\ -L & r_x^2 \end{pmatrix}. \tag{4.3}$$

Its eigenvalue at P_- is $J = -e^{-2\pi\mathrm{i}\delta}$, as follows from its lower row, and the other eigenvalue is J^{-1}. Thus the same pairing is

$$C = I + \frac{R_0 - 1}{J - J^{-1}}(P - J^{-1}I), \qquad CA_{\text{in}}C^{-1} = A_{\text{out}}, \quad CB_{\text{in}}C^{-1} = B_{\text{out}}. \tag{4.4}$$

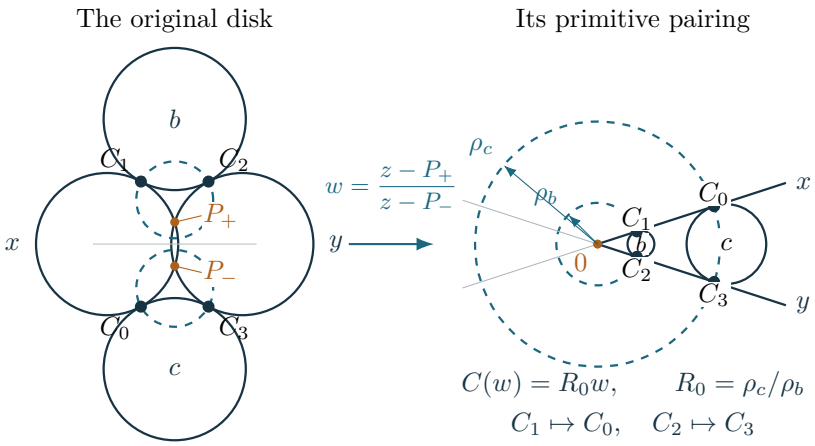


Figure 1: The two sides of one crossing disk, with $x = y = b = c = 1,\ \delta = 1/10$. Dashed circles bound the shaded planes. The Möbius coordinate sends the outer intersections P_+, P_- to $0, \infty$. The positive dilation C sends the ordered contacts C_1, C_2 to C_0, C_3. The meridian follows the inverse path. All labels refer to the same original disk.

Its eigenvalues on the selected and other lines are $R_0, 1$. If

$$c_{\mathrm{sc}} = \frac{\pi(T_b(0) + T_c(0))}{A} > 0,$$

then $R_0 = 1 + 2c_{\mathrm{sc}}\delta + O(\delta^2)$, and

$$C(0) = I + \frac{c_{\mathrm{sc}}}{2\pi \mathrm{i}}(P + I). \tag{4.5}$$

This retains the nontrivial parabolic limit.

4.2 The marked lines and the side signs

An independent local choice of exterior disk could change a side sign, so we justify the line limit using the whole marked pattern. Apply to every circle the single map

$$g_t(z) = \frac{z}{1 - tz + \mathrm{i}t/2}.$$

Its pole $1/t + \mathrm{i}/2$ lies between the marked halfplanes and outside all finite disks for small real t. All transformed outward disks are bounded. The marked curvatures are positive of order t^2; the others tend to their positive values. If a middle center is on side $\sigma_f = \pm 1$ of its outer center line, the preceding calculation gives the *signed* ratio

$$R_{\mathrm{signed}} = (E_b/\sqrt{D})^{\sigma_b}(E_c/\sqrt{D})^{-\sigma_c}. \tag{4.6}$$

The center triangles of bounded disjoint disks have one planar cyclic orientation, except for the outside triangle. Here the outside triangle is one of the two gaps containing both marked lines: every gap with only one line has bounded horizontal extent. Such a triangle has a middle and an outer curvature zero, hence $T_f = 0,\ E_f/\sqrt{D} = 1$ in the line limit.

For clarity we calibrate the nonzero terms by the complete cusp itself. Inversion at the outer tangency puts the outer planes at distance $W = A/2$. The inverted input and output middle circles have imaginary center coordinates $-T_b(0)/2$ and $T_c(0)/2$; thus the shaded height is $H = (T_b(0) + T_c(0))/2$. For example, before inversion the input center is

$$\frac{x - y}{bA} + \mathrm{i}\frac{2T_b(0)}{bA}, \qquad |\mathrm{center}|^2 - r_b^2 = \frac{4}{bA},$$

which gives these coordinates directly. The disk longitude traverses two widths, and the pure meridian one height. With the ordered paths of (4.1), their ratio is $2\mathrm{i}W/H$. One may check both the factor and the orientation using rectangle corners $(C_0, C_1, C_2, C_3) = (1, E, -1, -E)$, $E = e^{2\pi \mathrm{i} x_0}$, $0 < x_0 < 1/2$: the lower and upper white maps are

$$B_0(z) = -z/E + (E+1)/E, \qquad B_2(z) = -z/E - (E+1)/E.$$

The two cross-half disk maps replace $E+1$ by $E-1$. Their elementary paths give $\lambda = -2(E+1)$, $\mu_0 = E-1$; the signed half-collar meridians are -2 and $2E$. The parallelograms have the area of the two original rectangles, so the curves are primitive. In particular the pure sidepairing is

$$C(0) = I - \frac{\mathrm{i}}{2c_{\mathrm{geom}}}(P+I), \qquad c_{\mathrm{geom}} = W/H = \frac{A}{T_b(0) + T_c(0)}.$$

This computation uses complete rectangle maps, without presupposing the deformed ratio.

Conjugate (4.6) back by g_t^{-1} and let $t \to 0$. Its coefficient of $P+I$ becomes $-\mathrm{i}(\sigma_b T_b(0) - \sigma_c T_c(0))/(2A)$. The complete rectangle forces $\sigma_b = 1$ for a nonzero input term and $\sigma_c = -1$ for a nonzero output term. A zero term is irrelevant. These signs persist for small real δ, since a nondegenerate center triangle has squared height $4c_0^2 T_f^2/(f^2 D) > 0$. The exceptional line triangle has $T_f = 0$ throughout. The maps themselves have a regular limit: the distinct triple $(C_1, C_2, \text{selected fixed point})$ is sent to the distinct prescribed triple on the other side. The unique three-point Möbius map depends holomorphically on both triples. Thus (4.4), including both full generator relations, holds in the common marked pattern and extends to complex parameters.

4.3 Threading, half twists, and the whole diagram

The positive crossing convention on an ordered pair is

$$(A, B) \longmapsto (A^{-1}BA, A). \tag{4.7}$$

When A, B have the same trace and $P = BA$, this is simultaneous conjugation by $H = P^{-1} - I$. To verify it, substitute $A^2 - (\operatorname{tr} A)A + I = 0$ and the corresponding relation for B into $HA - (A^{-1}BA)H$ and $HB - AH$. Here $\det H = 2 - \operatorname{tr} P$ is a unit near completeness. Its selected eigenvalue ratio is $(J^{-1} - 1)/(J - 1) = -J^{-1}$. The same quadratic relation gives $H^2 = (\operatorname{tr} P - 2)P^{-1}$, so two positive crossings act by conjugation with P^{-1}.

The auxiliary circle is the literal three-strand threading word $[2, 1, 1, 2]$. Apply (4.7) chronologically to (A, B, C_{aux}). If $K = P^{-1} C_{\mathrm{aux}} P$, the outputs are

$$(K^{-1}AK,\ K^{-1}BK,\ K).$$

Closing the auxiliary strand gives $[C_{\mathrm{aux}}, P] = I$, and its pure transport is conjugation by C_{aux}^{-1}. Following the two underpasses along the disk gives $P = BA$; the auxiliary strand has no self crossing, so there is no meridian correction to this disk longitude. A retained crossing changes the transport by the specified power of H. Hence its original meridian eigenvalue $m = e^{U/2}$ and longitude logarithm are

$$m^{-2} = R_0(-J)^{\epsilon}, \qquad U = -\operatorname{Log} R_0 - \epsilon \operatorname{Log}(-J), \qquad V = 2\operatorname{Log}(-J) = -4\pi\mathrm{i}\delta. \tag{4.8}$$

An explicit determinant-one meridian matrix is

$$mR_0(J^{-1} - 1)^{-\epsilon} C^{-1}(P^{-1} - I)^{\epsilon}. \tag{4.9}$$

Its selected eigenvalue is m, and its determinant is $m^2 R_0(-J)^{\epsilon} = 1$. Negative half twists are included because all inverted factors are units. This also fixes the central signs of these

local representatives. The longitude is centered by replacing P with $-P$ when taking its small logarithm. This does not change its projective peripheral word, and is not a claim that every centered representative belongs to one global SL_2 lift. The similarity coordinates are twice these centered eigenvalue logs. Cutting the actual disk collar before a meridian filling also checks the surgery sign. Killing $\mu + p\lambda$ sets the auxiliary word to P^{-p}. Its inverse threading transport is P^p, which is the $-2p$ crossing twist by the preceding formula for H^2. Equivalently, a Dehn twist of the twice-punctured disk induces that same Hurwitz move on its ordered based paths. With the retained ϵ crossings the original box power is therefore $\epsilon - 2p$, as in (1.1); no outside strand or basepath is reconnected.

We now check that the local maps assemble for arbitrary cycles in the diagram, not just for a tree. Complexify a white reflection as a pair $\mathcal{R}_f^+, \mathcal{R}_f^-$ of mutually inverse maps between the two projection copies. On an actual sweep whose region sequence is $F_0, \dots, F_k$, set

$$Q_j = \mathcal{R}_{F_j}^+ \mathcal{R}_{F_0}^-, \qquad H_j = \mathcal{R}_{F_j}^+ \mathcal{R}_{F_{j-1}}^-.$$

Then $H_j \cdots H_1 = Q_j$. These are the original prefix frames. At a main tangency the raw product has trace -2; the positively oriented lift is its negative. Unchanged strands retain the same regions and frames. A cup or cap inserts or deletes F, G, F, and

$$\mathcal{R}_F^+ \mathcal{R}_G^- \mathcal{R}_G^+ \mathcal{R}_F^- = I.$$

Thus a change in the number of swept strands leaves every outside prefix unchanged. At a crossing disk (4.4) intertwines both ordered puncture generators, and the threading and half-twist words are exactly those just calculated. These equations are the Wirtinger and disk van Kampen relations of the original inverse-cut diagram. They construct a representation of $\pi_1(Y_D)$, with all its peripheral words, rather than a collection of trace matches. At zero it is the complete representation from Section 2.

Every main meridian remains a noncentral parabolic. Its commuting longitude is therefore a translation in the same fixed-point chart; its multiplier is one. The two main cusp translations remain real-linearly independent near the complete point. This preserves the complete main cusp and the original marking.

5 The classical critical point for independently signed fillings

The identity $E_b(\delta)E_b(-\delta) = D$ shows that Q_b is even in its own parameter. The common face-critical germ preserves this property. At zero its channel Hessian is diagonal:

$$\partial_e \partial_f A_0(0) = -2c_e\, \delta_{ef}, \qquad c_e = \frac{\pi(T_{b_e}(0) + T_{c_e}(0))}{x_e + y_e} > 0. \tag{5.1}$$

Indeed the first channel derivative is zero along the entire complete face chart, so face elimination adds no Schur correction to this block; differentiating (1.4) gives the displayed value. At least one of each pair's triangles is nonmonomial, since its matching edge cannot be the protected strand edge.

Separate evenness gives holomorphic functions B_e, real at real arguments, such that

$$\partial_e A_0 = -2\delta_e B_e(\delta_1^2, \dots, \delta_R^2), \qquad B_e(0) = c_e.$$

Shrink one fixed polydisk until $\operatorname{Re} B_e > 0$ throughout. Put $t_e = 1/n_e$. The critical equations are

$$t_e \partial_e A_0 + 2\pi \mathrm{i}\delta_e + 2\pi \mathrm{i} t_e = 0.$$

Their derivative in δ at $(t,\delta)=(0,0)$ is $2\pi\mathrm{i}I$, so they have a unique holomorphic solution for all small independent t_e. At a real nonzero t_e,

$$\delta_e = -\frac{t_e}{1+\mathrm{i}t_eB_e/\pi}, \qquad \operatorname{Im}\delta_e = \frac{t_e^2\operatorname{Re}B_e}{\pi|1+\mathrm{i}t_eB_e/\pi|^2} > 0. \tag{5.2}$$

The values B_e in this formula are at the joint critical point; they need not be real there. This argument handles either sign of each filling power without a quantum estimate.

Let $A_\epsilon = A_0 + \pi\mathrm{i}\sum_e \epsilon_e\delta_e^2$. Equations (1.4) and (4.8) give the exact one-form identity

$$dA_\epsilon = \sum_e U_e\,d\delta_e, \qquad U_e + p_eV_e = \partial_eS_{\mathbf{n}} - 2\pi\mathrm{i}. \tag{5.3}$$

Thus the critical point has the required primitive logarithmic period, and $\operatorname{Re}V_e = 4\pi\operatorname{Im}\delta_e > 0$. It tends to the complete representation as $\min_e|n_e|\to\infty$. What remains is to prove smooth metric completion and the absolute, rather than differential, volume equality.

6 From the logarithmic equation to smooth filling

We prove the metric implication in a form that states its hypotheses explicitly. This step does not require a triangulation of the filled manifold.

Proposition 6.1 (Near-complete primitive completion)**.** *Let M be an oriented complete finite-volume hyperbolic three-manifold with torus cusps and holonomy ρ_0. Fix actual primitive bases (μ_i,λ_i). There is a neighborhood of ρ_0, with its peripheral fixed points, in which the following holds. Let ρ be an actual representation, let U_i, V_i be its small similarity logarithms in bounded frames sending the selected fixed points to infinity, and suppose:*

(a) both multipliers are one at each cusp to be left unfilled;

(b) at each cusp to be filled, $a_iU_i + b_iV_i = \pm 2\pi\mathrm{i}$, where $\gcd(a_i,b_i)=1$.

Then ρ descends to the indicated original topological filling and is the holonomy of a smooth, complete, oriented hyperbolic metric of finite volume on it. The neighborhood is chosen before the slopes. This assertion supplies completion, not existence of ρ.

Proof. Identify the universal cover of M with $\mathbb{H}^3$. On a compact truncated core and a larger compact collar choose a finite evenly covered coordinate cover with lifted partition $\eta_{j,\gamma}$, where $\gamma\in\pi_1M$. In the future unit hyperboloid define

$$Y_\rho(x) = \sum_{j,\gamma}\eta_{j,\gamma}(x)\rho(\gamma)\rho_0(\gamma)^{-1}x, \qquad E_\rho(x) = \frac{Y_\rho(x)}{\sqrt{-\langle Y_\rho(x),Y_\rho(x)\rangle}}. \tag{6.1}$$

Positive convex combinations of future timelike vectors are future timelike. Reindexing the sum proves exact equivariance. At ρ_0 this is the identity; only finitely many terms occur on a compact fundamental part. Hence it is C^1-close to the identity and is a positive local diffeomorphism near ρ_0. No properness of the deformed group action has been assumed.

For a cusp, write the complete cover as $(w,t)\in\mathbb{C}\times[t_0,\infty)$, with translation lattice Λ and developing map (w,e^t). In its moving bounded frame the deformed action is $A_\lambda(z)=a_\lambda z+b_\lambda$. Choose a compactly supported smooth partition $\sum_{\lambda\in\Lambda}\eta(w-\lambda)=1$, and set

$$d_\rho(w) = \sum_\lambda \eta(w-\lambda)A_\lambda(w-\lambda). \tag{6.2}$$

Then $d_\rho(w+\nu) = A_\nu d_\rho(w)$. On a closed fundamental parallelogram it is C^1-close to w; its positive real Jacobian is nonzero. Equivariance gives the same property everywhere. Let $\chi : \mathbb{C} \to \mathbb{R}$ be the real-linear extension of $\lambda \mapsto \log|a_\lambda|$. The entire cusp map

$$(w,t) \longmapsto (d_\rho(w), e^{t+\chi(w)}) \tag{6.3}$$

is equivariant and has positive triangular Jacobian at every finite point. On a compact collar it is close to the complete map. Interpolate it with (6.1) along hyperbolic geodesics, using an invariant cutoff. The interpolation stays C^1-close on that compact collar. Thus we have an equivariant positive local diffeomorphism on the entire ideal-end-deleted manifold, and its pullback metric has curvature -1 without edge singularities.

At a filled cusp some multiplier differs from one. The commuting affine maps then have a common second fixed point $q = b_\lambda/(1-a_\lambda)$. Both complete generator translations are nonzero. Therefore $|q| \to \infty$, uniformly as $\rho \to \rho_0$ off the translation locus, by choosing either generator whose multiplier differs from one. The entire developed fundamental parallelogram and fixed generator paths stay in a bounded set and have bounded lengths. For a sufficiently small neighborhood their images miss q. Equivariance extends this exclusion over the whole link cover.

Choose $f = \operatorname{Log}(d_\rho - q)$ on that simply connected cover. The metric $|df|^2$ is nonsingular and descends to the compact torus. It is complete. Its lifted local isometry f covers $\mathbb{C}$: completeness lifts every finite Euclidean line segment. Since $\mathbb{C}$ is simply connected, f is a diffeomorphism and its deck periods form a rank-two lattice. For a fixed generator path with developed length at most L and image in $|d_\rho| \le R$,

$$|\Delta f| \le \frac{L}{|q|-R} \longrightarrow 0.$$

The exponentials of these increments equal e^U, e^V. Their differences from the small chosen logarithms are in $2\pi\mathrm{i}\mathbb{Z}$; smallness proves, exactly,

$$\Delta_\mu f = U, \qquad \Delta_\lambda f = V. \tag{6.4}$$

This is proved for the fixed basis before multiplying by the possibly large filling integers. It is the step that excludes an extra winding.

Translate q to zero. The function $e^{t_0+\chi(w)}/|d_\rho(w)-q|$ is positive and periodic, so is bounded above. The cusp map therefore contains a full epigraph $y > A|d_\rho - q|$. With $r = y/|d_\rho - q|$, put

$$R = \operatorname{arsinh}(1/r), \quad \theta = \operatorname{Im} f, \quad X = \operatorname{Re} f + \tfrac{1}{2}\log(1+r^2).$$

Direct substitution in the upper-half-space metric gives

$$ds^2 = dR^2 + \cosh^2 R\, dX^2 + \sinh^2 R\, d\theta^2. \tag{6.5}$$

The epigraph contains a whole punctured tube $0 < R < R_0$. A period L translates (X,θ) by $(\operatorname{Re} L, \operatorname{Im} L)$. By (6.4), the original primitive filling slope is exactly one angular turn. A complementary primitive period has nonzero real part, because the two periods have real rank two. Adding the axis $R = 0$ is therefore an ordinary smooth solid torus, with that precise meridian. The slope holonomy fixes q, ∞ and has multiplier one, so it is the identity; the representation descends to the filled group. The sign of the angular turn reverses the parameterization, not the orientation of the hyperbolic metric.

At an unfilled cusp the action is by translations. The same compact Euclidean-link argument shows that (6.3) supplies a complete horoball quotient. These ends have finite

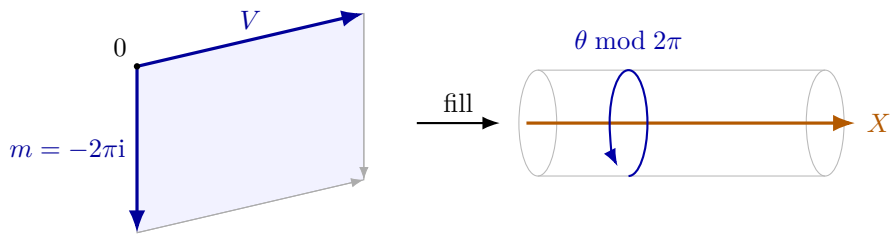


Figure 2: The logarithmic link and its filling. The vertical period $m = U + pV = -2\pi\mathrm{i}$ is the primitive meridian of the added solid torus; the other displayed period is the original disk longitude V. The old basis is recovered by $U = m - pV$. The tube coordinates (X, θ) identify translation by V with a translation along the axis together with a rotation, while m is one full angular turn. The drawing uses a scaled vertical axis.

volume. The middle is compact and the added tubes have finite volume; the whole metric is complete. Its universal developing local isometry covers the simply connected space $\mathbb{H}^3$, by complete geodesic lifting. It is an isometry. This proves discreteness and faithfulness for the filled topology as well. □

The branch in Section 5 satisfies every hypothesis of Proposition 6.1. It is an actual representation in one bounded complete frame; the selected flags are holomorphic; the main peripheral groups remain translations; and $\operatorname{Re} V_e > 0$. The exact slope equation is $-2\pi\mathrm{i}$. This proves the metric assertion of Theorem 1.1, including zero restrictions on relative signs or relative rates of the independently long powers.

7 The absolute action and the same filled fundamental chain

7.1 The peripheral differential and its factor

Fix the shape convention before computing a sign. We orient upper half-space by $dx \wedge dy_{\mathrm{hor}} \wedge dh$. The positively ordered model is $(a_0, a_1, a_2, a_3) = (0, \infty, 1, z)$, with $\operatorname{Im} z > 0$; approximating infinity by height H gives determinant $H \operatorname{Im} z > 0$. For general vertices our cross-ratio is

$$z = \frac{(a_3 - a_0)(a_2 - a_1)}{(a_3 - a_1)(a_2 - a_0)}.$$

Its signed volume is $D(z)$. The outward cusp orientation at infinity is the usual complex orientation. We take peripheral bases of positive intersection in that orientation, as is the rectangle basis with $\operatorname{Im}(\lambda/\mu) > 0$.

Retain the positive ideal triangulation of the original exterior constructed in Section 2. Develop its ideal vertices at the continued peripheral flags of the representation. Their ordered cross-ratios remain positive near the complete point. Every face and edge equation holds because the developed vertices come from the same representation and the original group words. Let $\beta_\rho = \sum_\Delta [z_\Delta]$, with the original orientations, be this fundamental ideal chain, and put $\mathcal{V}_\rho = \sum_\Delta D(z_\Delta)$. Here the Bloch–Wigner volume is

$$D(z) = \operatorname{Im} \operatorname{Li}_2(z) + \log|z| \arg(1 - z), \qquad dD = \log|z|\, d\arg(1 - z) - \log|1 - z|\, d\arg z.$$

The local logarithms are continued in this one positive shape chart.

We spell out the boundary normalization used below. For a positively ordered tetrahedron use logarithms $u = \log z$ and $v = \log(1 - z^{-1})$, with symplectic form $du \wedge dv$; reverse the form for a reversed tetrahedron. The three opposite-edge coordinate vectors are

$$e_0 = (1, 0), \quad e_1 = (0, 1), \quad e_2 = (-1, -1).$$

The type of edge ij is 0 for $01, 23$, 1 for $02, 13$, and 2 for $03, 12$; thus $\mathrm{type}(j, f)$ means the type of edge joining vertices j, f. Face f is opposite vertex f. Here is the geometric identification of the normal-path rows. At each vertex send that vertex to infinity and normalize its first two remaining vertices, in increasing order, to $0, 1$. The third is respectively $z^{-1}, z, (1 - z^{-1})^{-1}, (1 - z)^{-1}$ for vertices $0, 1, 2, 3$. The logarithms of the three opposite cusp-side vectors, in increasing other-face order, are therefore, up to constant multiples of πi,

j	ℓ_{f_0}	ℓ_{f_1}	ℓ_{f_2}
0	v	$-u$	0
1	$u + v$	u	0
2	$-u - v$	$-v$	0
3	$-v$	$-u - v$	0.

(7.1)

For instance at infinity, $j = 1$, the cusp triangle is $0, 1, z$ and its sides have vectors $z - 1, z, 1$. If a_i is the scale taking the i-th canonical cusp triangle into a common developed chart, a shared side gives $a_{i+1}/a_i = -\mathrm{side}_{\mathrm{out}}(i)/\mathrm{side}_{\mathrm{in}}(i + 1)$. Around a closed normal path the scales telescope. Its actual similarity multiplier is the product of outgoing sides divided by incoming sides, including the fixed signs. Thus its logarithmic row is the sum of the outgoing-flow-weighted entries of (7.1). The table proves that this row is exactly the vector $d_j(r)$ below, in the coordinates (u, v).

The following local calculation records the intersection factor. In tetrahedron orientation ε, at vertex j list the other faces $f_0 < f_1 < f_2$. A normal flow has outgoing entries $r_{j,f}$ across the three sides of that cusp triangle, with $\sum_{f \neq j} r_{j,f} = 0$. In the next two formulas we suppress the first index within a fixed vertex j, writing $r_{f_i} = r_{j,f_i}$. Set

$$d_j(r) = (-1)^j \left(r_{f_1} e_{\mathrm{type}(j,f_2)} - r_{f_2} e_{\mathrm{type}(j,f_1)} \right), \qquad \tau_j = -\varepsilon(-1)^j.$$

The oriented triangle intersection is $\frac{1}{2}\tau_j(r_{f_0} s_{f_1} - r_{f_1} s_{f_0})$. Expanding the two-coordinate determinants gives

$$\Omega_\varepsilon(d(r), d(s)) + \sum_j \tau_j(r_{f_0} s_{f_1} - r_{f_1} s_{f_0}) = \sum_f \varepsilon(-1)^f r_f^{\mathsf{t}} \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} s_f. \tag{7.2}$$

Here $d = \sum_j d_j$, and the column r_f on the right lists $(r_{j,f})_{j \neq f}$ in increasing vertex order, as does s_f. This identity is verified by substituting the three displayed vectors and eliminating the third flow at each vertex. Each flow therefore has eight independent variables; expanding all their bilinear coefficients proves the identity. The face terms cancel on glued faces: a permutation matrix P conjugates the displayed skew matrix to $\mathrm{sign}(P)$ times itself. Thus peripheral rows pair by minus twice their oriented intersection number. Distinct tori pair to zero.

For completeness, a small normal circle about an endpoint of a global edge has vector equal, up to orientation, to that edge's signed shape-incidence row: each successive corner contributes its opposite-edge type. It is null-homologous in the cusp triangulation. The same face cancellation in (7.2) therefore makes every edge row orthogonal to every closed peripheral row and to every other such small edge circle. This proves edge isotropy. Its only

linear dependencies are the cusp-vertex ones. A dependency assigns numbers λ_{ij} to the six edges of each tetrahedron whose opposite sums are equal. It has the form $\lambda_{ij} = s_i + s_j$: solve for the four s_i from any three incident pairs and the opposite-sum equations. A face identifies the three s_i uniquely, so they descend exactly to one value per ideal vertex. For t tetrahedra and c torus cusps, the number of edges is t, by Euler characteristic, hence the edge rank is $t - c$. The row quotient has dimension $2c$. Primitive peripheral rows form a basis because their pairings are -2 on each cusp and zero between cusps. To pass from rows to tangent vectors, let $J = \bigoplus_\Delta \varepsilon_\Delta \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and let E be the edge-row span. The edge-equation tangent space is the annihilator $W = \{X : aX = 0 \ (a \in E)\}$; its null directions are JE^{t}. Peripheral rows annihilate these null directions. If a, b are the meridian and longitude rows of one cusp, then $aJb^{\mathsf{t}} = -2$. Modulo null directions their dual tangent vectors are $-Jb^{\mathsf{t}}/2$ and $Ja^{\mathsf{t}}/2$. Their J-pairing is $-1/2$. Cross-cusp pairings vanish, and the $2c$ vectors exhaust the quotient W/JE^{t}. Thus its tangent form is

$$-\frac{1}{2}\sum_i dU_i \wedge dV_i.$$

In these shape coordinates, $dD = \operatorname{Re} u \, d \operatorname{Im} v - \operatorname{Re} v \, d \operatorname{Im} u$. The real shape-log vector and the variation of its imaginary part both lie in W: the real edge sums are zero and the imaginary edge sums are constant multiples of 2π. Their peripheral coordinates are respectively $(\operatorname{Re} U, \operatorname{Re} V)$ and $(d \operatorname{Im} U, d \operatorname{Im} V)$. Applying the reduced pairing to these two vectors proves

$$d\mathcal{V}_\rho = \frac{1}{2}\sum_i (\operatorname{Re} V_i \, d \operatorname{Im} U_i - \operatorname{Re} U_i \, d \operatorname{Im} V_i). \tag{7.3}$$

The fixed central signs in peripheral lifts have zero real part and zero derivative, so centering their logarithms at completeness does not alter this calculation. This supplies the factor for similarity logs, rather than eigenvalue logs. The classical peripheral-volume framework is discussed by Neumann–Zagier [4]; the calculation above fixes the present conventions component by component.

7.2 Absolute calibration and the filling correction

Equation (5.3) gives

$$d(2\pi \mathrm{i} A_\epsilon) = -\tfrac{1}{2}\sum_e U_e \, dV_e.$$

Taking real and imaginary parts and using (7.3) shows that $2\pi \operatorname{Re} A_\epsilon - \mathcal{V}_\rho + \frac{1}{2}\sum_e \operatorname{Im} U_e \operatorname{Re} V_e$ is constant. It vanishes at completeness by (2.4)–(2.5). Hence

$$2\pi \operatorname{Re} A_\epsilon = \mathcal{V}_\rho - \frac{1}{2}\sum_e \operatorname{Im} U_e \ \operatorname{Re} V_e. \tag{7.4}$$

This uses the complete volume of the original polyhedra as the constant. Agreement of derivatives or Hessians would not fix it.

At the critical point, $U_e = 4\pi \mathrm{i} p_e \delta_e - 2\pi \mathrm{i}$ and $V_e = -4\pi \mathrm{i} \delta_e$. Directly,

$$\operatorname{Re}(-2\pi \mathrm{i} p_e \delta_e^2 + 2\pi \mathrm{i} \delta_e) = \frac{1}{4\pi} \operatorname{Im} U_e \ \operatorname{Re} V_e. \tag{7.5}$$

Since $S_{\mathbf{n}} = A_\epsilon + \sum_e (-2\pi \mathrm{i} p_e \delta_e^2 + 2\pi \mathrm{i} \delta_e)$, (7.4) becomes $2\pi \operatorname{Re} S_{\mathbf{n}} = \mathcal{V}_\rho$.

7.3 Why this is the volume of the filled metric

The last equality must use the same original fundamental class. We give the comparison explicitly. For a peripheral map fixing an axis, use a comparison map taking the end into that axis. After a conjugation its endpoints are $0, \infty$, and it has the form $(0, Ye^{\chi(w)})$, where the character χ is real-linear on the peripheral cover. The original conical developing chain has the form $(a(w), Yb(w))$ on each boundary triangle; on a finite fundamental set a, b and their derivatives are bounded and $b > 0$. Use precisely that boundary triangulation for the homotopy

$$H_s(w, Y) = \big(sa(w),\, Y \exp\big((1-s)\chi(w) + s \log b(w)\big)\big), \qquad 0 \le s \le 1. \tag{7.6}$$

It is equivariant because both heights transform by the same positive multiplier and the horizontal coordinate by the same complex multiplier. On a fundamental boundary triangle the height is comparable to Y, horizontal derivatives are bounded, and height derivatives are $O(Y)$. Hyperbolic volume is $y^{-3}\, dx\, dy_{\mathrm{hor}}\, dy$; therefore the homotopy's three-dimensional volume is $O(Y^{-2})$. The same estimate applies to any subdivision and to the parabolic conical comparison at a main cusp.

After filling, extend this *same* axis-valued boundary triangulation over the solid torus. The primitive slope has been killed, so the extension is equivariant; the remaining cyclic group acts by translation along the axis. Every added tetrahedron is collinear and has zero signed volume. Stokes, first on finite truncations and then as $Y \to \infty$, now identifies the original ideal-chain volume $\mathcal{V}_\rho$ with the equivariant volume of the filled representation. Replacing an unrelated new set of vertices by axis points would not establish this equality.

Finally compare this equivariant chain with the developing map of the complete metric from Proposition 6.1. They represent the same filled fundamental class and agree at the remaining cusp flags. Straightening their common finite subdivision and applying the same vanishing boundary estimate gives $\mathcal{V}_\rho = \mathrm{Vol}$ of that metric. This proves (1.8) for the constructed complete metric. The usual hyperbolic volume is independent of a choice of complete metric; the following short volume argument also establishes the needed independence without a rigidity equality case.

Lemma 7.1. *Two oriented complete finite-volume hyperbolic metrics on the same marked cusped three-manifold have the same volume.*

Proof. We recall the inequality needed for this conclusion. For a complete source metric and a boundary-parabolic representation, choose an equivariant map, affine-conical on its cusps. It is globally Lipschitz after a compact-core smoothing. Smear a finite regular hyperbolic tetrahedron T_R over the oriented isometry quotient, with Haar measure normalized by the source volume. Fubini gives the integral of its pulled-back volume as $\mathrm{Vol}(T_R)\,\mathrm{Vol}(\rho)/\,\mathrm{Vol}(M)$. Straighten the four image vertices. The difference is the sum of the four face-homotopy prisms. Their integrals cancel in pairs: the ordered faces of a regular tetrahedron are congruent, Haar measure is invariant, and their boundary signs alternate. The Lipschitz bound on fixed finite T_R justifies these finite integrals; cusp truncations give the same formula by the preceding boundary estimate.

Every straight tetrahedron has absolute volume at most the regular ideal volume v_3. In the Klein ball enlarge its vertex rays to the ideal boundary; an oriented nondegenerate tetrahedron's unsigned volume increases. For an ideal tetrahedron the Lobachevsky formula is $\Lambda(\alpha) + \Lambda(\beta) + \Lambda(\gamma)$, with $\alpha + \beta + \gamma = \pi$. It vanishes on the boundary. An interior critical point has $\sin\alpha = \sin\beta = \sin\gamma$, hence $\alpha = \beta = \gamma = \pi/3$. Its maximum is therefore v_3. Thus

$$|\,\mathrm{Vol}(\rho)| \le \frac{v_3}{\mathrm{Vol}(T_R)}\,\mathrm{Vol}(M).$$

Let $R \to \infty$. Apply the resulting inequality to the holonomy of each metric with the other as source. Its equivariant volume is its metric volume by the same fundamental-chain comparison. The two opposite inequalities prove equality. □

8 The interface with source asymptotics

Theorem 1.1 has only classical inputs: the labeled matched sphere, the original disk paths, and the independently long fixed integer fillings. In particular, the functions $Q_b, A_0, S_{\mathbf{n}}$ were defined here by angles, not by a limit of polynomials. If a finite source calculation produces these same local functions and the original primitive framing, it may use the following precise output.

Corollary 8.1 (Geometric identification of a selected action)**.** *Suppose a separately proved source theorem for the specified original diagram has the action* (1.6)*, with its roots and logarithms continued from the complete face-critical point. For fixed independently sufficiently long signed powers, its unit-charge critical point is the classical point of Section 5. Its real limiting action is* $\mathrm{Vol}(S^3 \setminus K_{\mathbf{p}})/(2\pi)$ *when the original component test is a knot, and the corresponding filled link volume otherwise.*

Proof. The source equations and this paper's equations are identical in the specified germ. The implicit-function uniqueness in Section 5 identifies their critical points. Theorem 1.1 then gives the claim. □

The companion oscillatory filling theorem [6, Theorem 1.1] requires a nonzero uniform local packet and a strict complementary strip bound for the *actual finite polynomial*. The present paper neither assumes nor proves those analytic hypotheses. It also gives no comparison between central and one-sided crossing sectors. To deduce a full invariant limit, the finite diagram identity, all parity and framing factors, noncancellation of the whole central period, and a bound on the remaining original sectors must be supplied separately. The geometric conclusion identifies their volume target when those inputs have been proved; it does not replace them.

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